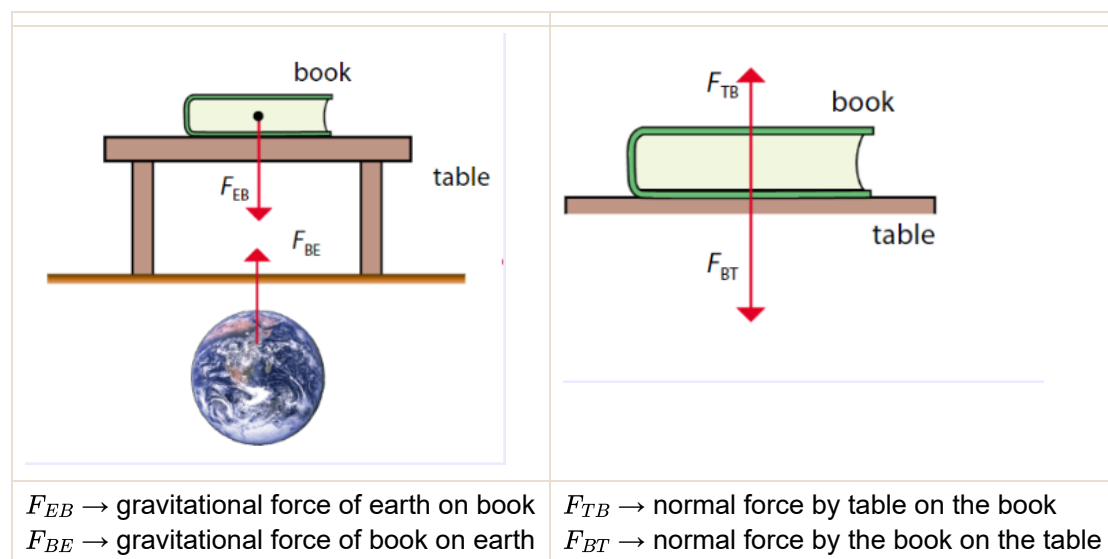
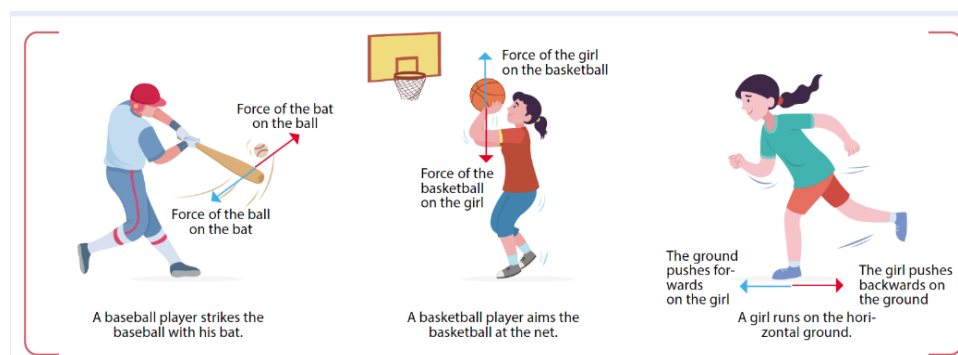


Newton's Third Law

Definition: Newton's third law of motion states that if body A exerts a F_{AB} on body B, then body B will exert an equal and opposite force F_{BA} on body A.

#PhysicsDefinitions



Pair of action reaction forces are shown

- gravitational force $F_{EB} \rightarrow$ exerted by earth on book $\rightarrow$ and the upward force F_{BE} exerted by book on earth
- Normal force $F_{TB} \rightarrow$ by table on the book $\rightarrow$ and the normal force F_{BT} by the book on the table

Each action and reaction pair are equal $\Rightarrow$ in magnitude but act in opposite directions.

Hence $\Rightarrow$ force of the table on the book $\Rightarrow$ does not form an action reaction pair with the weight of the book.

Newton's third law of motion tells us 4 characteristics of force

1. forces always occur in pairs $\Rightarrow$ each pair is made up of an action and a reaction force
 2. action and reaction forces are equal in magnitude
 3. action and reaction forces act in opposite directions
 4. action and reaction forces act on different bodies
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How are 2 forces in an action reaction pair identical

- both forces are of same type
- magnitudes of both forces are equal

How are 2 forces in an action reaction pair different

- Act in opposite directions
 - Act on 2 different bodies
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